

No DIYian is an Island

Like open source software, DIY is also greatly driven by the community—both online and physical. In fact, it is the community spirit that has made DIY so tangible today

"I think the biggest positive aspect is the community that has grown around DIY projects. It is not so much the individual projects being generated by this new DIY resurgence that are particularly novel. People have always pursued their passions, tinkered and innovated unique solutions to their everyday problems. What is currently so powerful about DIY projects is that these people are no longer operating alone. Even if you are the only person in your town who does this sort of thing, you can go online and connect with people all over the world who think like you do. Collectively this community has pushed innovation much further than people ever could have working alone in their basement," says Randy Sarafan of Instructables.

Being part of a community helps DIYers to collaborate with like-minded individuals to build their projects faster and better. Sometimes, it also becomes possible to combine modules into a bigger solution. Moreover, since larger DIY solutions are often multidisciplinary, collaboration is indispensable since one person rarely has all the skills required for the fruition of an idea.

While online communities are a great boon and beneficial to people living in remote areas, it also helps to have physical gatherings if there is a DIY group in your town. Sayan Chakraborty, who is working towards setting up a physical hackerspace for IoTBLR, says, "Meeting and working in person is very important for the DIY maker/hacker community as we are mostly dealing

with physical things, whether it is electronic devices or artwork or product designs. In addition, some of the skills and capabilities being acquired and exchanged between the community members require active participation and discussion while interacting with physical devices at the same time."

The IoT Bangalore, Computer Club of India, Jaaga and HasGeek are some of the groups that are quite active in Bengaluru. These groups run regular group buys for components or make runs to the local electronics component market. In addition, there are constant barbers or donations to specific member projects.

"Our meetings are very informal and in the spirit of the makerspace/hackerspace culture we actively encourage collaborative learning. Interesting project ideas and implementations are discussed openly and voted on. Teams are formed based on the interest of the members. One of the key concepts we are dealing with these days is the evolving 802.15.4 wireless standard. We have had a session on the pros and cons of open standards (like 802.15.4) vs proprietary wireless standards. We have also had a very fruitful session on hardware platforms ranging from the well-known Arduino, MSP430 and Raspberry Pi to the relatively less-known but extremely energy-efficient EFM32. We all come from different fields and these sessions enable us to learn new things and share ideas."

ahead of the yesteryears.' (See box 'No DIYian is an Island'.)

The result

Great products and often thriving companies are born of DIY projects. There is no dearth of examples in this space.

Sharing his favourite Pi-based projects, Upton says, "The high-altitude ballooning that Dave Akerman has been doing in the UK (<http://www.daveakerman.com/>) remains my favourite, but there are lots of others, from the Siri-activated garage door opener (<http://www.raspberrypi.org/phpBB3/viewtopic.php?t=63&t=25118&p=231345>) to the Raspberry Pi bartender ([\[ics/bartendro-a-cocktail-dispensing-robot\]\(http://www.kickstarter.com/projects/partyrobot-ics/bartendro-a-cocktail-dispensing-robot\)\). What's great is that many of these projects start off as DIY efforts and then turn into small-scale businesses \(for example, <http://www.kickstarter.com/projects/tubecore/tubecore-duo>, which just got funded\)."](http://www.kickstarter.com/projects/partyrobot-</p>
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Kuber shares, "There is a new device called ArduSat. It's an Arduino Satellite wherein Arduino users can test their code in space (<http://blog.arduino.cc/2013/08/12/ardusat-successfully-launched-in-space/>). It is exciting to know that now Indian students would be able to have the same access to space! ArduSat has tied up with Dhruva Space India and we are partnering with them to teach space-oriented experiments in a very

fun and educative way (http://www.spaceschool.co.in/?page_id=561)."

Regarding popular projects in his group, Chakraborty says, "There are a number of interesting projects and we encourage people to form their own teams based on their interest or capabilities. Some of today's popular themes include automation for home, retail, medical, agriculture and location tracking. My current favourite is a pollution monitoring and reporting system that uses GPS location tracking and mesh networking between the nodes."

Guarding against pitfalls

The DIY scene is generally charged with enthusiasm and positivity. Despite that, DIYers need to be aware of the potential pitfalls and how to guard their project against them.

Denmead warns, "The biggest pitfall is perhaps that sometimes doing a project yourself may not be the most economical way to achieve it."

"In India, DIY is more or less confined to academic interests and hence it is highly constrained by the goals and timelines. Another aspect is how

Online learning resources

- Electronics For You (www.electronicsforu.com)
- Make Magazine (<http://makezine.com/>)
- Instructables (<http://www.instructables.com/>)
- Arduino website (<http://arduino.cc/>)
- Raspberry Pi website (<http://www.raspberrypi.org/>)
- Beagleboard (<http://beagleboard.org/>)
- Popular Mechanics DIY (<http://www.popularmechanics.com/technology/how-to/>)
- DIY (<http://diys.org/>)
- Adafruit (<http://www.adafruit.com/>)